Project Title:
E-Commerce Product Entity-Resolution & Matching Platform

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The E-Commerce Product Entity-Resolution & Matching Platform is a machine learning-based system designed to identify and match identical or highly similar products listed across different e-commerce platforms. Online marketplaces often contain duplicate or inconsistent product listings due to variations in names, descriptions, brands, and pricing, making accurate product comparison difficult. This project applies data preprocessing, feature extraction, and similarity analysis techniques to compare product attributes and determine whether two listings refer to the same product. Machine learning algorithms are used to improve matching accuracy by learning patterns from labeled data. The system reduces duplicate records, enhances product search, supports price comparison, and improves customer experience by providing reliable product information. It also helps online retailers maintain clean and consistent product catalogs, leading to better inventory management and informed business decisions. Overall, the platform offers an efficient, scalable, and accurate solution for product entity resolution in modern e-commerce environments.

Signature of the Guide